FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

Analysis and Functional Architecture

SWAID - Plate Resonance (PR)

PLATE RESONANCE
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1 Introduction

1.1 Document Scope

This document presents Deliverable #4 for the Systems Engineering course and defines the analysis and functional architecture of the SWAID Plate Resonance (PR) subsystem. The scope covers system objectives, constraints, risks, functional decomposition, interface structure, and technology selection rationale for the current project phase.

1.2 Acronyms and Terms

SWAID Standing Wave Acoustic Interactive Demonstrator.

PR Plate Resonance subsystem of SWAID.

FDD Functional Decomposition Diagram.

N2 Interface mapping representation based on subsystem interaction flows.

FPGA Field-Programmable Gate Array.

DAC Digital-to-Analog Converter.

1.3 Document Revision

Table 1: Revision history

Version	Date	Author(s)	Description
1.0	2026-02-27	Lucas Aparício	Template creation
2.0	2026-03-19	Alexandre Gorra / Artur Kovbasa / Hugo Gomes	First Version
3.0	2026-03-20	Alexandre Gorra / Artur Kovbasa / Hugo Gomes	Final Version

2 Analysis

2.1 System Perspective and SMART Objectives

The SWAID Plate Resonance (PR) subsystem bridges the Human Interface (HI) and the physical Chladni Plate (CP). It interprets high-level commands via a predefined resonance dictionary to generate synchronized, multi-channel acoustic signals, forming stable cymatic patterns in a direct, open-loop architecture.

Objectives are defined as SMART targets:

- **O1 – Coverage**: Generate at least 8 predefined, visually distinguishable resonant patterns.
- **O2 – Latency**: Achieve system latency (command to acoustic output) $\leq$ **700 ms** in 95% of trials.
- **O3 – Stability**: Maintain each pattern for at least **3.0 s** without visible mode switching.
- **O4 – Repeatability**: Reproduce the target pattern in **9 out of 10** trials post-calibration.

A pattern is considered **stable** if its geometry remains consistent for the dwell interval under constant excitation.

2.2 Stakeholders and Operational Scenarios

Stakeholders include **Primary users** (interactors), **Operators** (setup/supervision), and the **Development team**. The runtime scenarios are:

1. **Initialization**: PR executes self-diagnostics, initializes ALSA drivers, loads the dictionary, and enters a "ready" state under 10 seconds.
2. **Normal Operation**: PR parses an HI command string, synthesizes the corresponding multi-frequency signal, and outputs it to the sound card.

2.3 Drivers, Constraints, and Risk Assessment

Engineering drivers include multi-channel capability, bounded latency, and repeatability. Constraints involve physical footprint, development time, and safe operating limits.

Table 2: Risk register for SWAID-PR

ID	Risk	Description	Level	Mitigation
R1	Resonance drift	Environmental or mounting variation shifts resonant frequencies during operation.	High	Pre-demo calibration and frequency retuning profile.
R2	Settling mismatch	Transition commands are issued before physical pattern stabilization.	Medium	Enforce minimum dwell and settling timers in controller.
R3	Interface mismatch	Inconsistent command semantics between interaction and PR control modules.	High	Freeze interface contract and perform integration tests.
R4	Channel imbalance	Uneven gain/phase across channels degrades pattern clarity.	Medium	Per-channel calibration and gain normalization.

2.4 Requirements Specification Baseline

Table 3 defines the refined and testable technical requirements for the PR subsystem.

Table 3: PR subsystem requirements baseline

ID	Type	Requirement Statement	Verification
PR-07	Safety	The PR subsystem shall digitally limit the maximum amplitude and duty cycle of the output signals to prevent exceeding the CP transducers' thermal ratings.	Inspection/Test
PR-08	Physical	The computational hardware of the PR subsystem must fit entirely within the structural footprint enclosure designated by the CP team.	Inspection
PR-09	Ext. Interface	The PR subsystem shall provide a software API to the HI team to accept high-level command strings (musical notes, chords, gestures).	Interface Test
PR-10	Int. Interface	The PR subsystem shall transmit actuation commands to the sound card using low-latency audio drivers (e.g., ALSA) for multi-channel synchronization.	Integration Test
PR-11	Reliability	The software/firmware shall support continuous operation for a minimum of 8 hours without requiring a system reboot or memory flush.	Endurance Test
PR-12	Performance	The subsystem shall automatically load the resonance dictionary and enter a "ready-to-receive" state within 10 seconds of system power-on.	Timed Test
PR-13	Latency	Total system latency from HI command reception to acoustic output generation shall not exceed 700 ms.	Timed Test

2.5 Technology Trade-Off Matrix

Table 4 summarizes the technology decision using weighted criteria aligned with project constraints. Scoring scale: 1 (poor) to 5 (excellent).

Table 4: Technology trade-off matrix for excitation strategy

Criterion	Weight	Analog	Sound Card	FPGA	Microcontroller
Channel count capability	0.25	1	4	5	2
Latency and timing control	0.25	2	4	5	3
Integration complexity	0.20	5	4	2	3
Development effort/time	0.20	5	4	1	3
Cost and availability	0.10	4	4	2	4
Weighted score	**1.00**	**2.95**	**4.00**	**3.25**	**2.85**

The **audio interface (Sound Card)** approach achieves the highest weighted score. It balances the need for multi-channel output (5.1/7.1 is sufficient for 4-6 transducers) with low implementation complexity, utilizing direct driver-level configurations (ALSA) to maintain the required latency margins.

3 Functional Architecture

3.1 Top-Level Functional Decomposition

The functional architecture maps the PR subsystem's pipeline, converting interactions into physical resonance purely through a calibrated, open-loop execution.

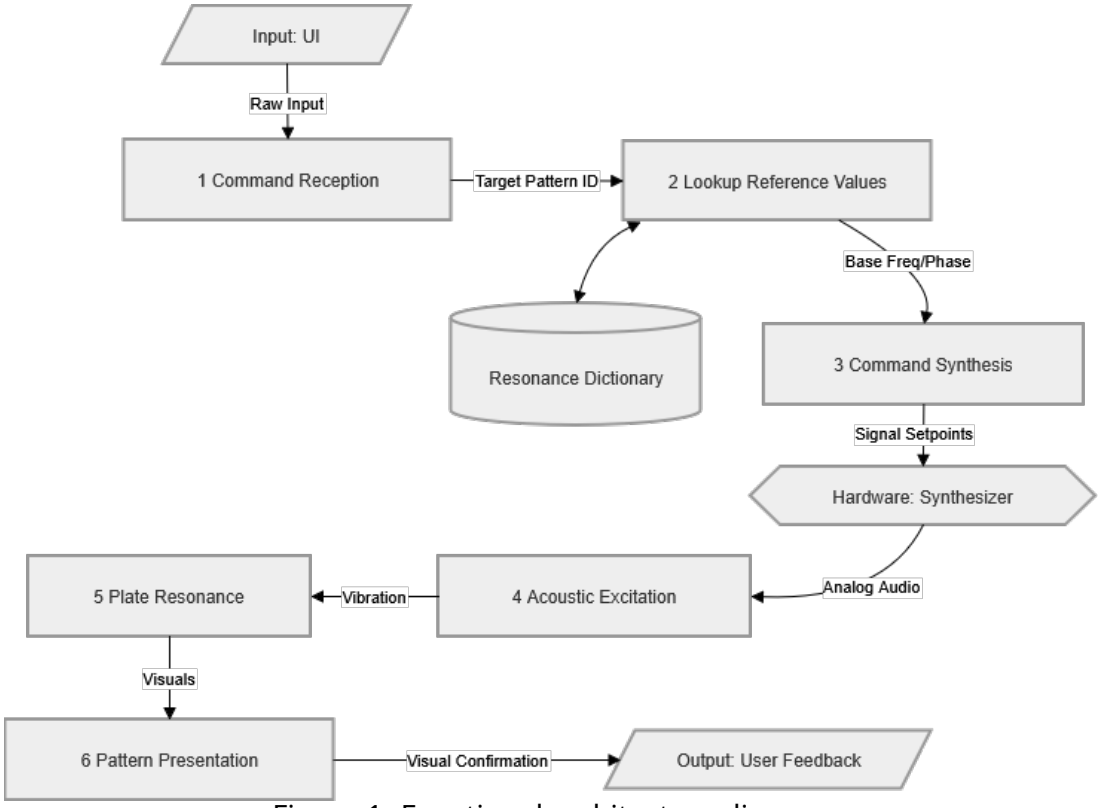

Figure 1: Functional architecture diagram

Core functional blocks include:

- **Command Interpretation:** Decodes external API strings from the HI subsystem.
- **Dictionary Lookup:** Correlates musical abstractions with empirical physical data (frequencies).
- **Signal Synthesis:** Generates discrete digital waveforms via software-based direct digital synthesis.
- **Output Protection:** Monitors duty cycle and digital gain to prevent hardware damage.

3.2 Functional Flow and Control Loops

The system generates patterns based on strictly pre-calibrated dictionary values. The architecture relies on a direct synthesis pipeline combined with a fast internal safety loop.

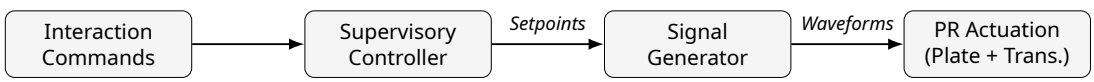

Figure 2: Open-loop control architecture for SWAID-PR with internal safety limiting

3.3 Function-to-Subsystem Allocation

Table 5 maps each identified function to its primary subsystem and interface outputs.

Table 5: Function-to-subsystem allocation for SWAID-PR

Function	Primary Subsystem	Outputs / Interfaces
Interaction acquisition	Human-interface / vision module	Symbol or command events to supervisor layer.
Intent interpretation	Supervisory control software	Target symbol identifier and transition request.
Excitation planning	PR control logic (Dictionary)	Frequency-amplitude-phase setpoints and dwell timing.
Signal generation	Low-latency audio driver (ALSA)	Synchronized digital audio streams to Sound Card.
Physical actuation	Transducers + Chladni plate	Mechanical vibration field and visible sand pattern.
Safety enforcement	Digital Limiter layer	Clamped output waveforms, shutdown/alert events.

3.4 Interface Architecture (N2 View)

Figure 3 provides an N2-style interface map of the principal subsystem interactions.

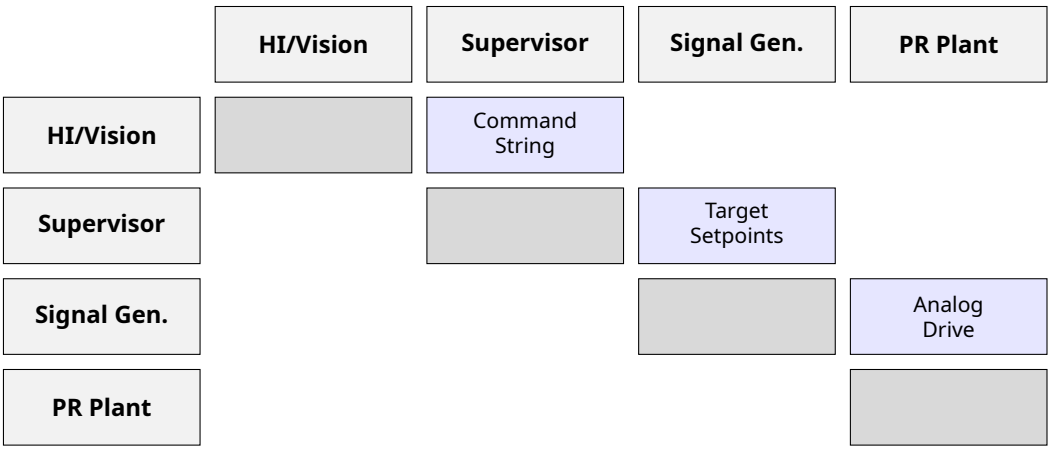

Figure 3: N2 interface diagram for SWAID-PR open-loop subsystems

The architecture separates command semantics from waveform generation, reducing cross-subsystem coupling and simplifying testing.

3.5 Verification-Oriented Functional View

Each major functional block is associated with a specific, verifiable outcome:

- **Command correctness**: The parsed command successfully maps to an existing entry in the loaded Dictionary.
- **Temporal behavior**: The system complies with the $\leq$ 700 ms latency requirement (Req. PR-13) and cold-boot under 10s (Req. PR-12).

- **Pattern stability**: The pattern remains physically stable without degradation for at least 3.0 s.
- **Safe operation**: Software limiters actively truncate signals attempting to exceed predefined thresholds (Req. PR-07).